

ECEN 474/704: Lab 05

Current Mirrors (using advanced layout design for both single stage current mirror and cascode current mirror)

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Due date: 2025-10-24

I. DISCUSSION OF THE LAB AND RESULTS

Include a short write-up (a paragraph or two) summarizing the lab and results. Compare the final designs with your pre-lab designs (as applicable for the lab), and make note of any major changes in your designs and why these changes were made. Regarding the simulation results, briefly describe any trends that are shown from the data or differences between pre- and post-layout simulation results, and attempt to provide an explanation for the trends and/or differences. Additionally, consider some of the following questions: were the results as you expected? Why or why not? What, if any, design trade-offs are demonstrated by the work in the given lab? If you were unable to meet any specifications given in the lab, what do you believe was the primary cause and what could be some potential solutions for the future?

The above questions are not an exhaustive list of what could be discussed and may not be relevant for all labs, but rather they serve as a starting point for you to demonstrate your understanding of the lab material.

This lab investigated the design, simulation, and layout effects of current mirror circuits (simple and cascode variants). The objective was to design mirrors that meet target output current ($120\text{ }\mu\text{A}$ for the simple mirror) and compliance voltage / output impedance specifications, verify functionality in schematic (pre-layout) simulation, perform layout extraction, and compare post-layout simulation results. During schematic design we sized devices to meet effective V_{eff} , expected g_m/I_{ds} and output resistance targets; during layout we prioritized symmetry, finger-count optimization, and guard/dummy devices to reduce mismatch and parasitic effects.

The comparison between pre-lab design targets and post-lab simulation results reveals expected deviations primarily caused by parasitic effects introduced during layout. For the **simple current mirror**, the pre-lab calculations predicted a compliance voltage of approximately **130 mV** and an output impedance around **49 k Ω** , which closely aligns with the post-layout results except for a slight increase in compliance voltage to **170 mV**. This small increase is attributed to parasitic resistances in the drain/source routing and contacts, which reduce the effective voltage headroom available at the output node. The output current and DC operating point ($V_o \approx 541\text{ mV}$) remained nearly identical before and after layout, confirming that the circuit's biasing and matching strategy were effective.

For the **cascode current mirror**, both pre- and post-layout simulations show a compliance voltage near **500 mV**, demonstrating that the stacked topology's headroom requirements dominated over parasitic effects. However, a minor drop in output impedance was observed — from **1.011 M Ω** pre-layout to approximately **972 k Ω** post-layout — likely due to the additional parasitic capacitances and channel-length modulation variations introduced by layout extraction. Despite this, the post-layout impedance still meets design expectations, validating the effectiveness of the cascode configuration in achieving high output resistance.

Overall, the post-layout results followed the expected trend: **slightly higher compliance voltages and marginal reductions in output impedance**, primarily due to layout parasitics and non-ideal routing effects. No major functional deviations were observed. The results confirm that the original pre-lab transistor sizing provided a solid foundation, requiring only layout-driven refinements such as increased finger count, dummy transistors, and symmetric placement to mitigate mismatch. The exercise highlights the inherent trade-off between **layout complexity and circuit precision**—achieving higher matching accuracy and output resistance comes at the cost of larger area and slightly increased voltage headroom.

II. SUMMARY OF RESULTS

Summarize the simulation results from the lab in one or two tables, like what is shown below.

Table 1

Example summary of results from pre-layout calculations

1.Simple current mirrors:

Parameters	V _{tn} mV	Lambda uv-1	ft	Uncox(ua/v2)	VGS	W	L	W'	L''
M1	447.0368	0.20997	34.1123	214.326	698.48	720n	270n	730n	270n
M2	447.0368	0.20997	34.11223	214.326	698.48	720n	270n	730n	270n

2.Cascode current mirror

Parameter	VEFF(n=2)	VGS	VDS	W	L	w'	L'	W''	Width of each finger	L''	Number of fingers
M1	0.1566	0.60363	0.1566	5.940u	270n	5.940u	270n	5.7u	570n	270n	10
M2	0.1566	0.60363	0.1566	11.931u	270n	11.931u	270n	10.26u	570n	270n	18
M3	0.3132	0.76023	0.447	1.408u	270n	2.08u	400n	2.28u	570n	270n	4
M4	0.3132	0.76023	0.3134	2.89u	270n	4.02u	400n	3.42u	570n	270n	6
MB	0.4698	0.91683	0.9168	0.574u	270n	0.574u	270n	0.570u	570n	270n	1

No of dummies :2

Total fingers-41

Table 2

Example summary table of pre- and post-layout results

1.Simple current mirror

Parameter	Compliance voltage	Vo @120	zout
Pre-layout	130mV	541mV	48.962K
Post-layout	170mV	541.6mV	48.962K

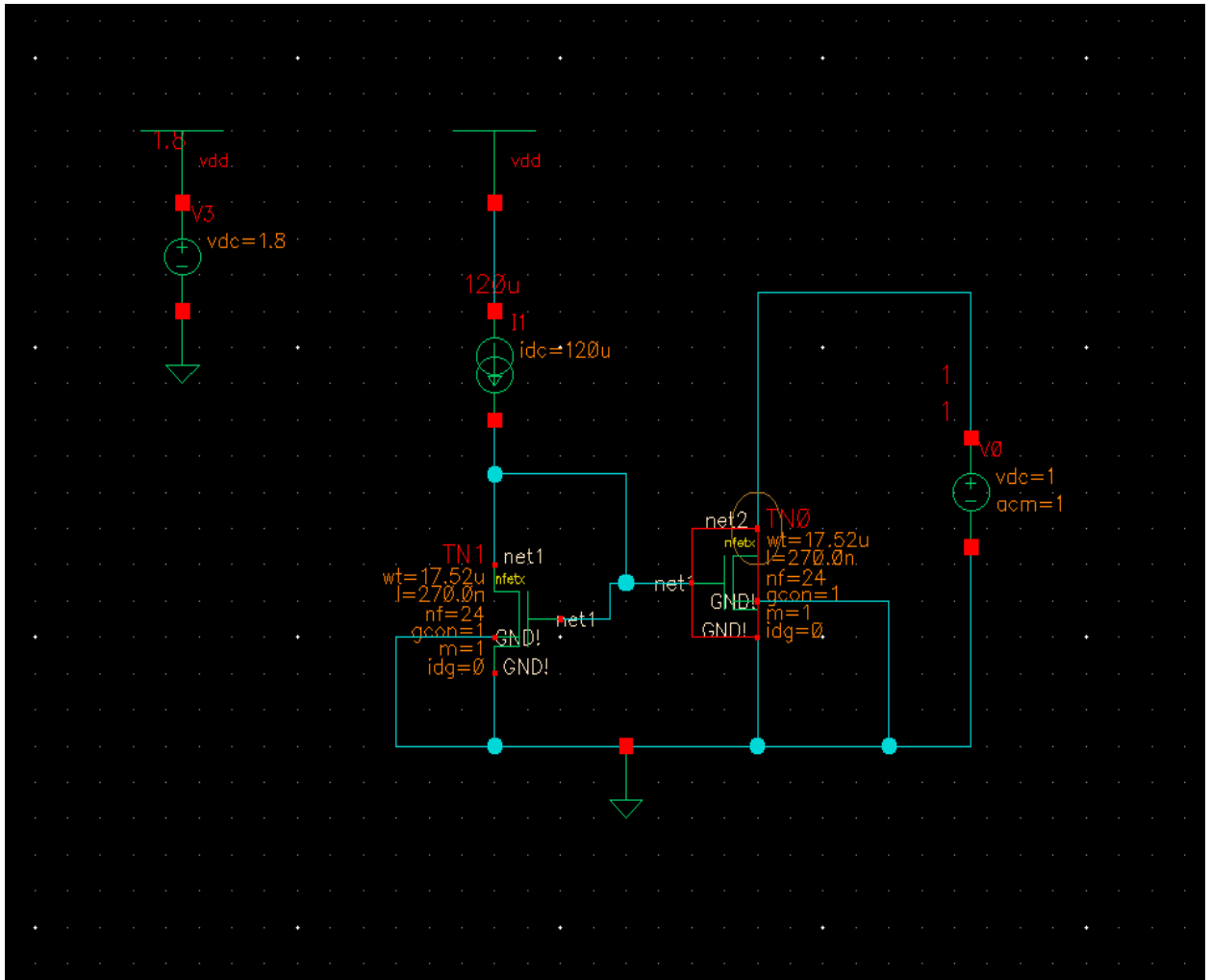
2.Cascode current mirror

Parameter	Compliance voltage	Vo @120	zout
Pre-layout	500mV	879.3433	1.011M
Post-layout	500mV	879.3433	972.38M

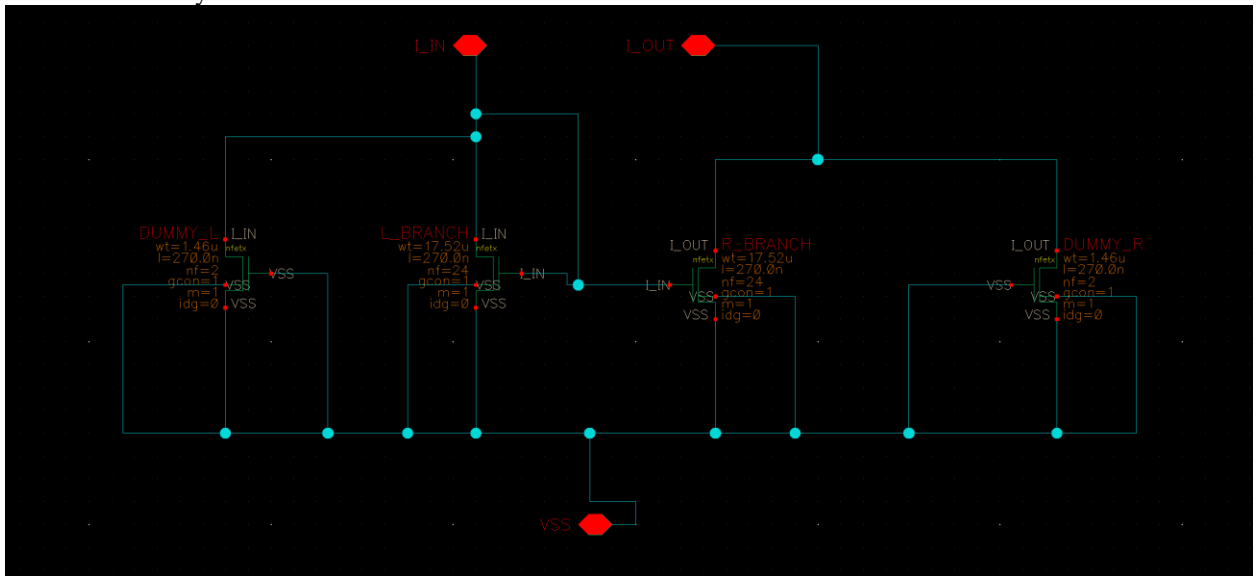
III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

Schematics of the sub-circuit(s) being designed in the lab should be included here. In addition, include images of the test bench schematics. Important signals in the test bench schematic should be labeled to provide clear correspondence between the signals in the schematics and the signals that are plotted in the simulation results shown later in the report.

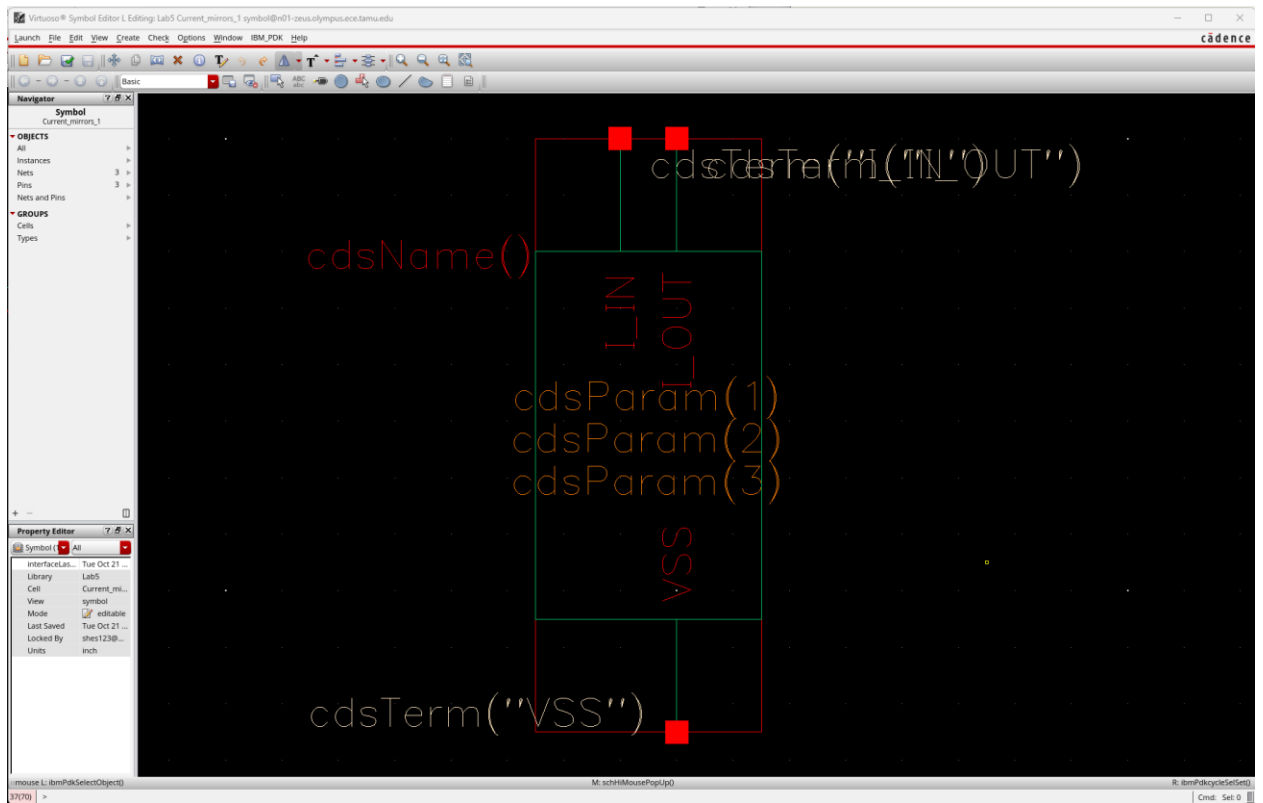
1.CIRCUIT FOR SINGLE STAGE:



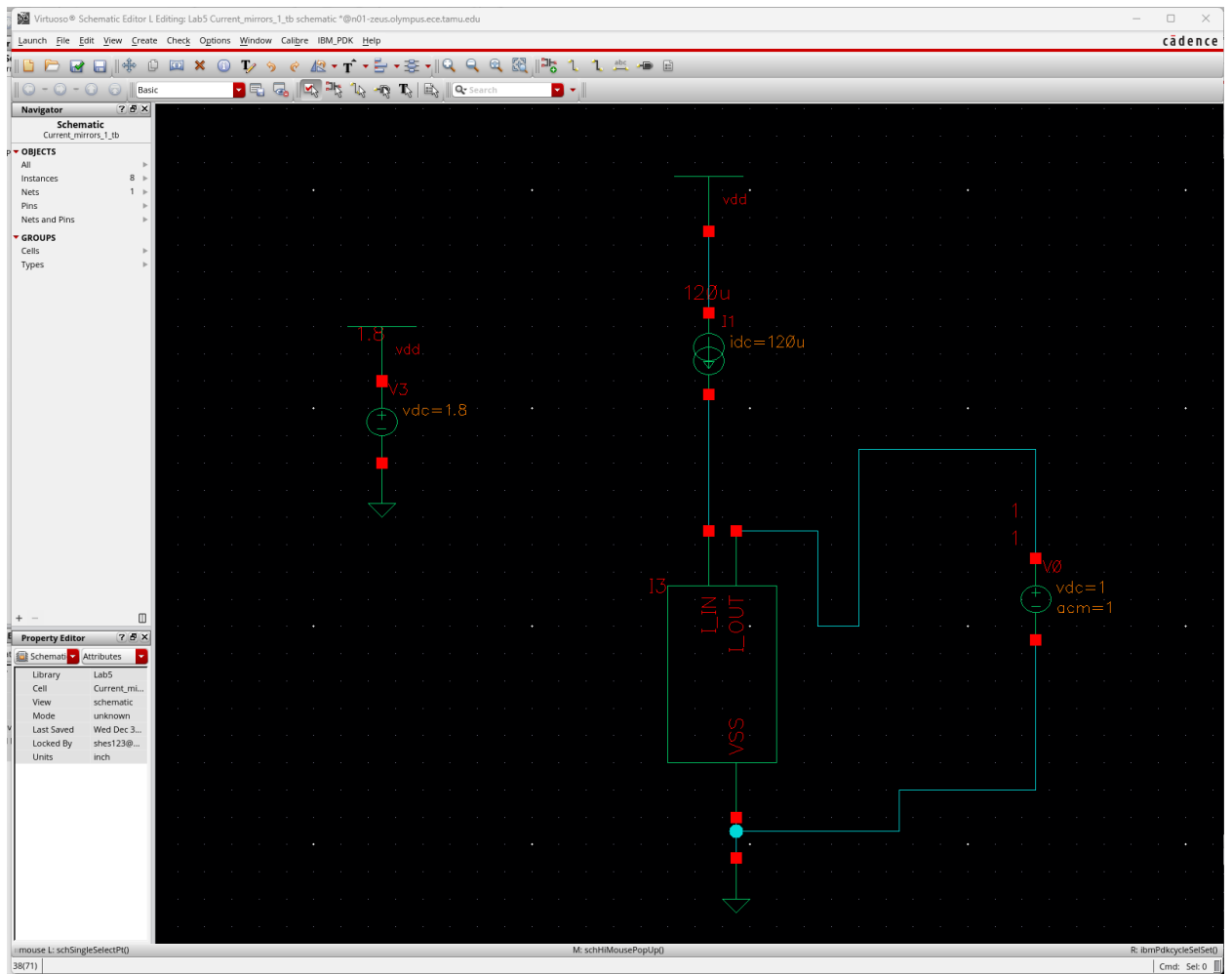
2.Schematic for Symbol creation



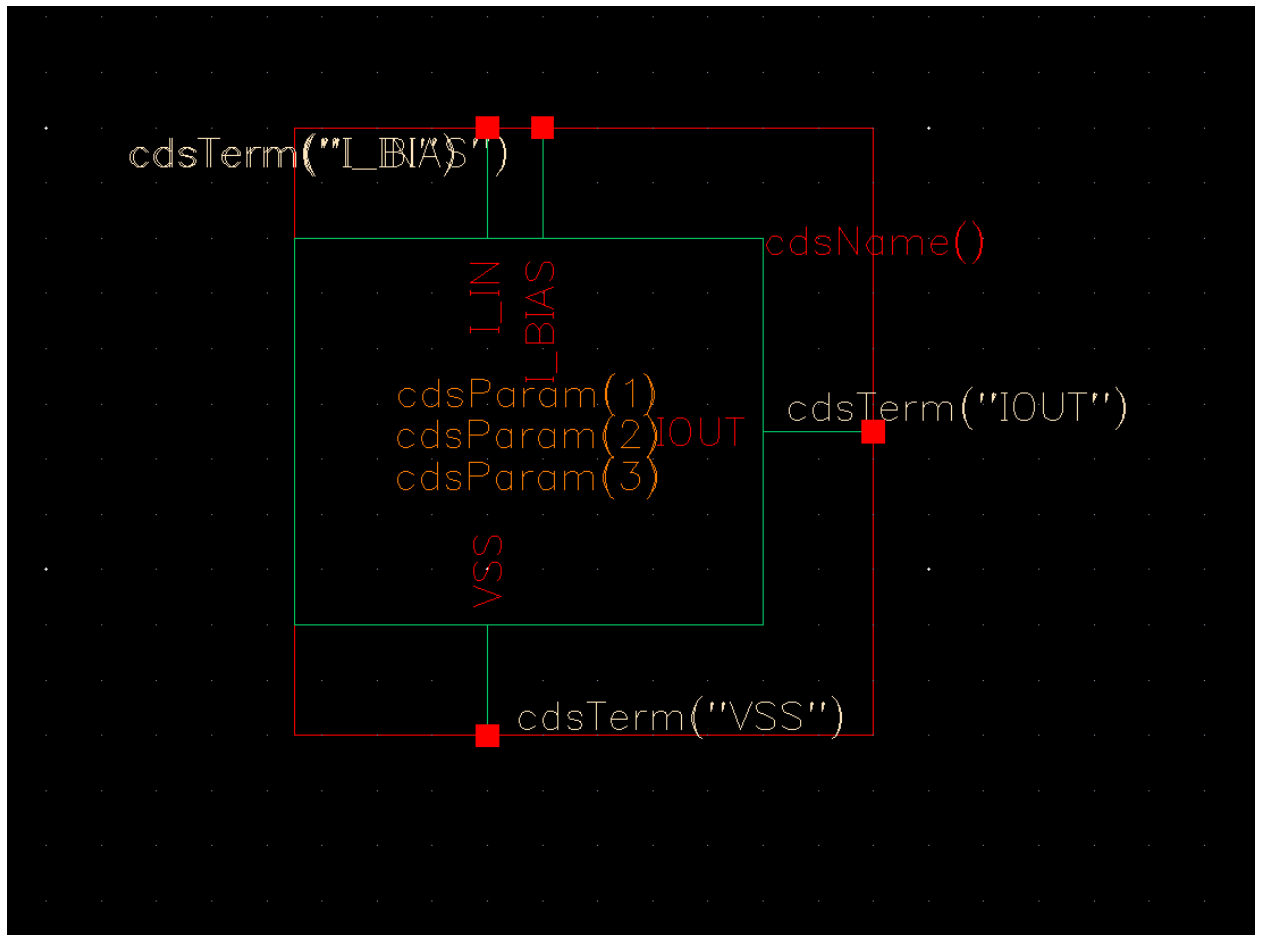
3.Symbol created



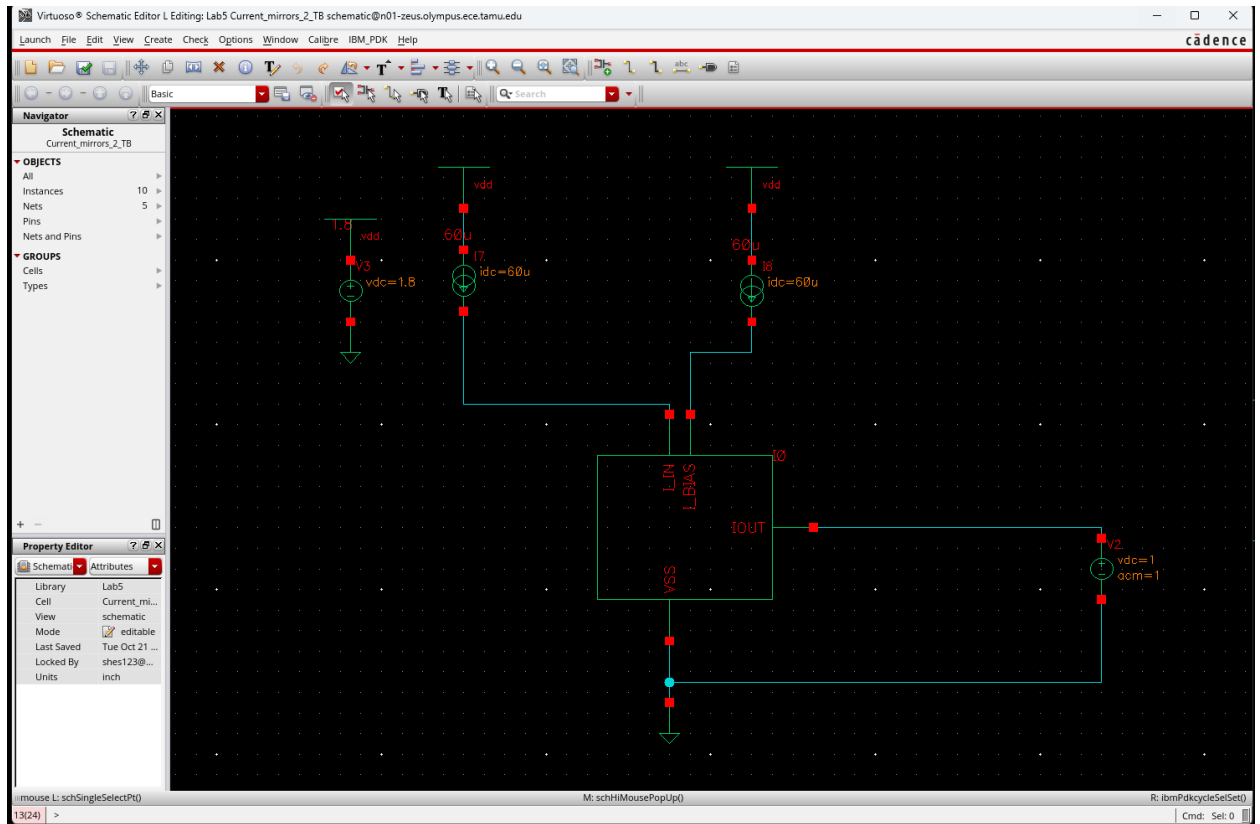
4. Actual test bench



2.CIRCUIT FOR CASCODE CURRENT MIRROR



Actual test bench used



IV. LAYOUT(S) AND FLOOR PLAN(S)

Include diagrams of the floor plans used to lay out major circuit components. Typically, this will just include transistor floor plans (routing does not need to be included in the diagrams). A table can be used to show the layout arrangement for the transistors or the floor plan can be drawn in another software tool. Clearly label the devices in the floor plan and make sure there is clear correspondence with the devices in the schematics, and include dummy devices in the floor plan diagram (dummy devices are often denoted with an asterisk, *).

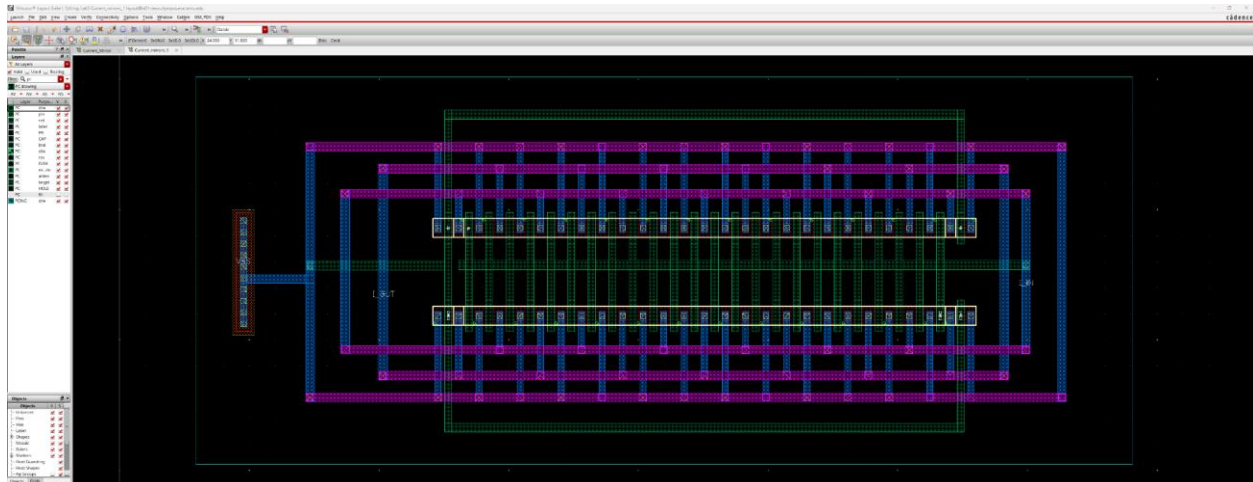
This section will also include images of the layout(s) of each circuit constructed for the lab.

1. SIMPLE CURRENT MIRROR

Floor plan:



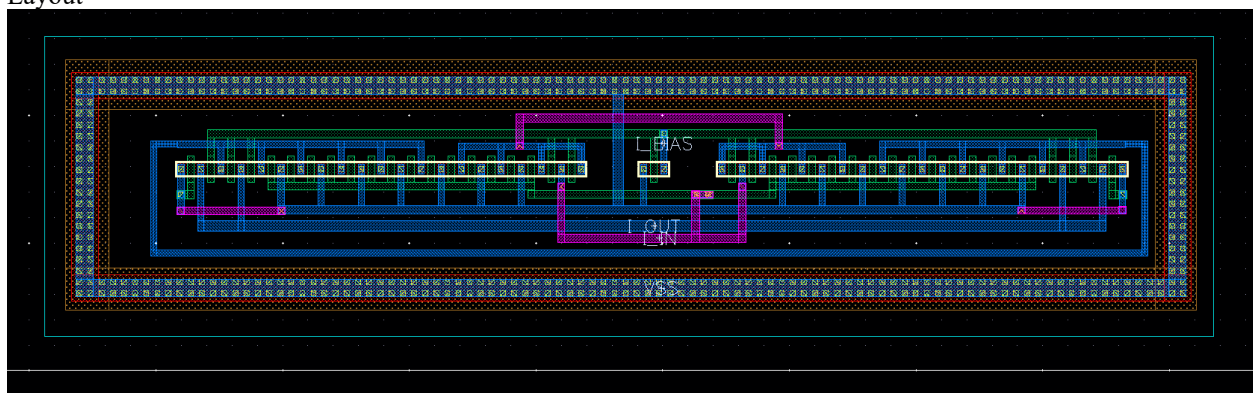
Layout:



2.Low Voltage Cascode Current Mirror
Floor plan:

D M4 M4 M4 M2 M2 M2 M2 M2 M2 M2 M2 M1 M1 M1 M1 M1 M3 M3 M MD M3 M3 M1 M1 M1 M1 M1 M2 M2 M2 M2 M2 M2 M2 M2 M4 M4 M4 D

Layout



V. DRC AND LVS RESULTS

This section should include the DRC RVE window results for each layout cell (with the results filtered to “show unresolved” to show only the remaining DRC violations), and the cell name should be visible at the top of the DRC RVE window. The LVS report summary image should also be included for each cell with your NetID, the time stamp, and the smiley face/check mark visible to show that LVS passed.

1.SIMPLE CURRENT MIRROR

[illegible]

Calibre - RVE v2024.2_29.16 : Current_mirrors_1.drc.results

FileViewHighlightToolsWindowSetupHelp

Filter: Show UnresolvedNo Results Found

Check / CellResults

Calibre Run Completed Successfully -- Results are Valid

Calibre Interactive - nmlVLS v2024.2_29.16 : /disk/amsr/IBM_PDK/cmnhv7sf/rel/Calibre/runset.lvs*@n01-zeus.olympus.ece.tamu.edu

FileSettingsConfigurationsHelp

RulesInputsH-CellsOutputsERCSignaturesRun ControlSearchTranscriptFiles

Current_mirrors_1.lvs.report

123456789101112131415161718192021222324252627282930313233343536373839404142434445464748495051

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##

CALIBRE SYSTEM

LVS REPORT

#####

REPORT FILE NAME: Current_mirrors_1.lvs.report

LAYOUT NAME: Current_mirrors_1.sp ('Current_mirrors_1')

SOURCE NAME: Current_mirrors_1.src.net ('Current_mirrors_1')

RULE FILE: _cmnhv7sf.lvs.cal_

CREATION TIME: Tue Oct 21 01:05:05 2025

CURRENT DIRECTORY: /home/grads/s/shes123/CAL

USER NAME: shes123

CALIBRE VERSION: v2024.2_29.16 Thu May 2 07:35:42 PDT 2024

OVERALL COMPARISON RESULTS

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##

CORRECT

#####

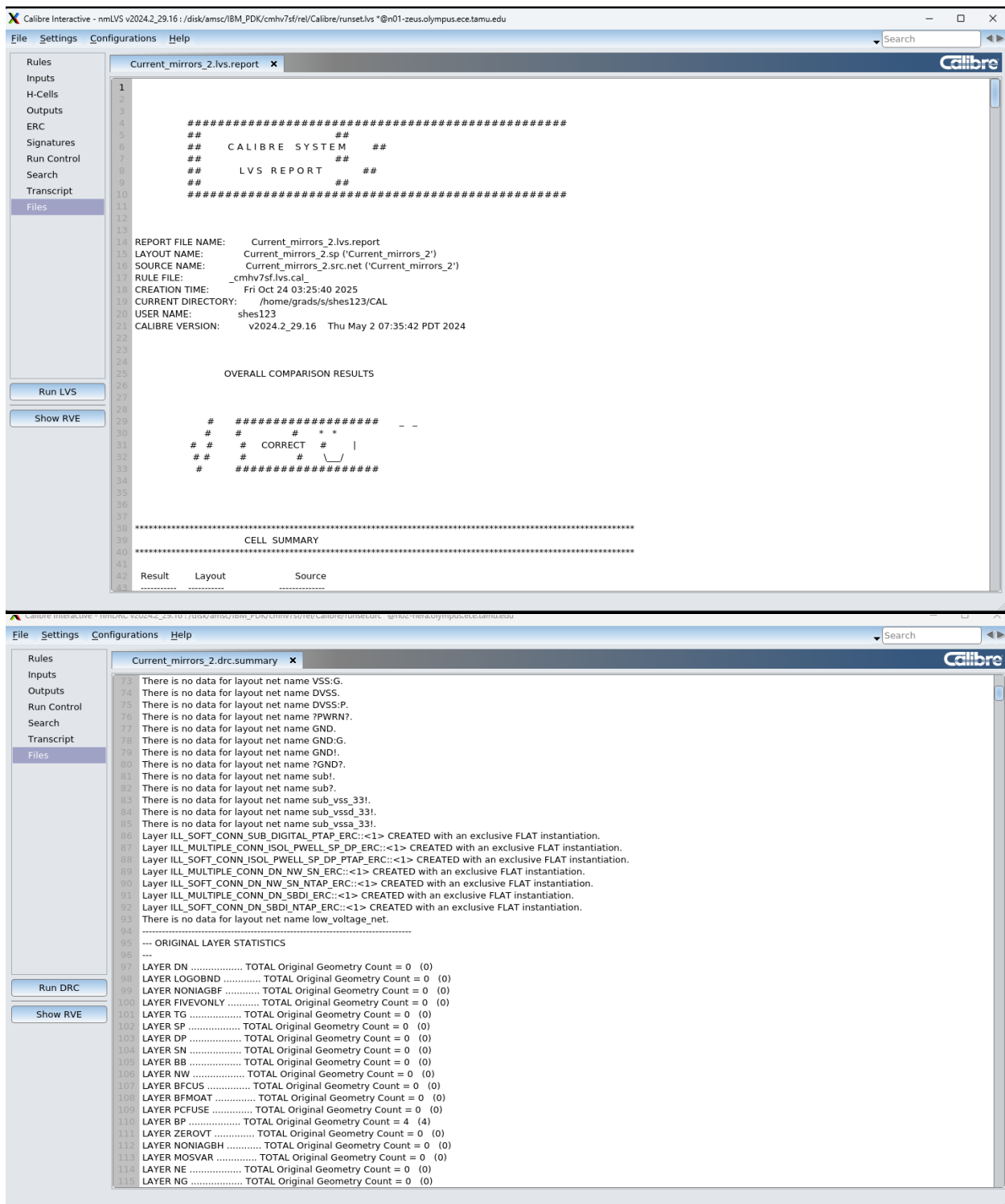
CELL SUMMARY

Result	Layout	Source
CORRECT	Current_mirrors_1	Current_mirrors_1

LVS PARAMETERS

2.LOW VOLTAGE CASCODE CURRENT MIRROR

Calibre Run Completed Successfully -- Results are V

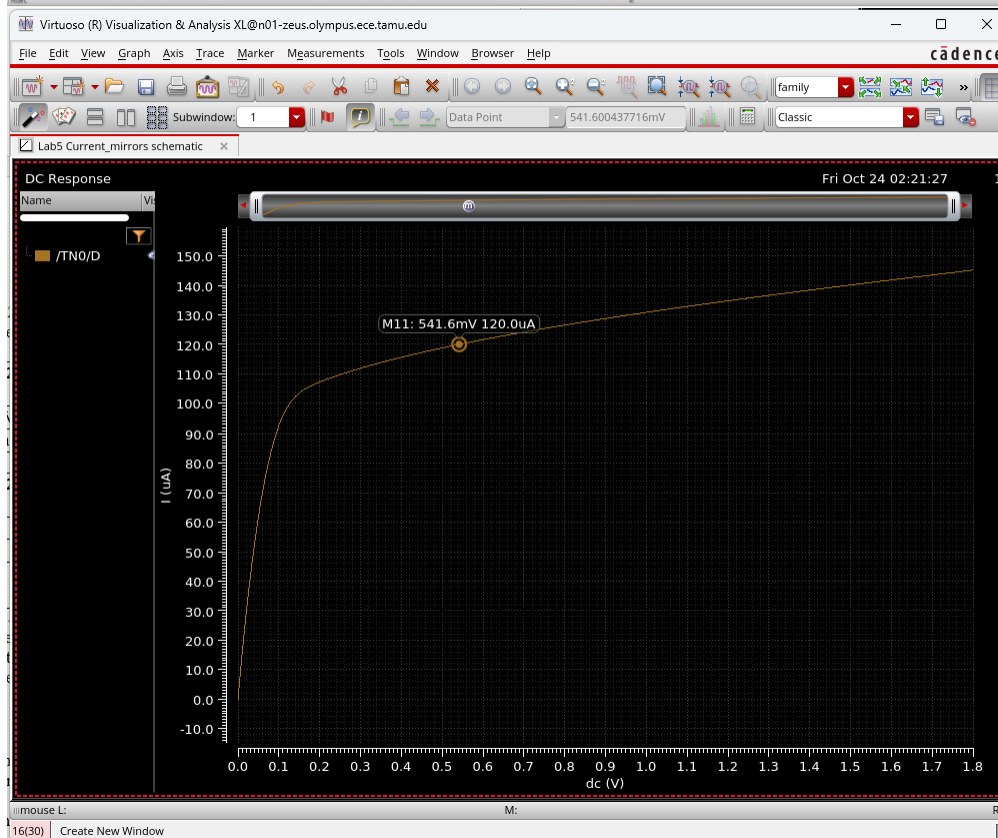
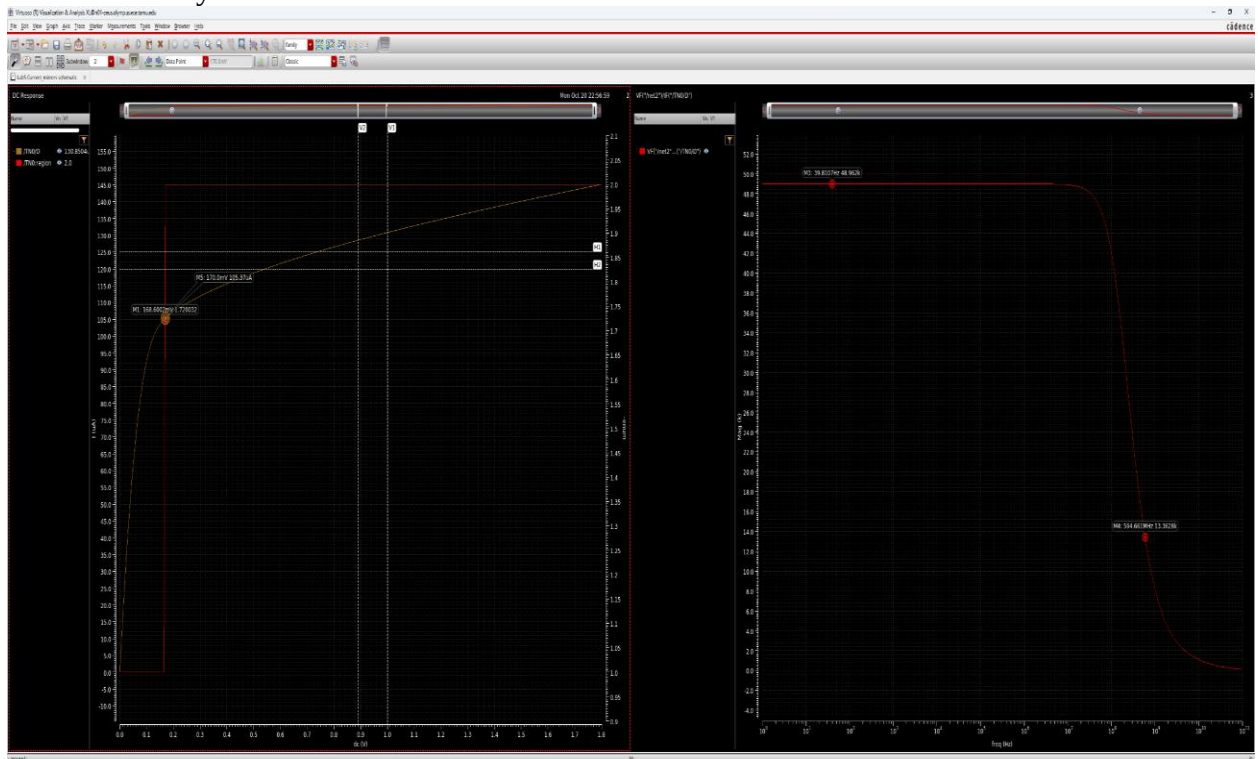


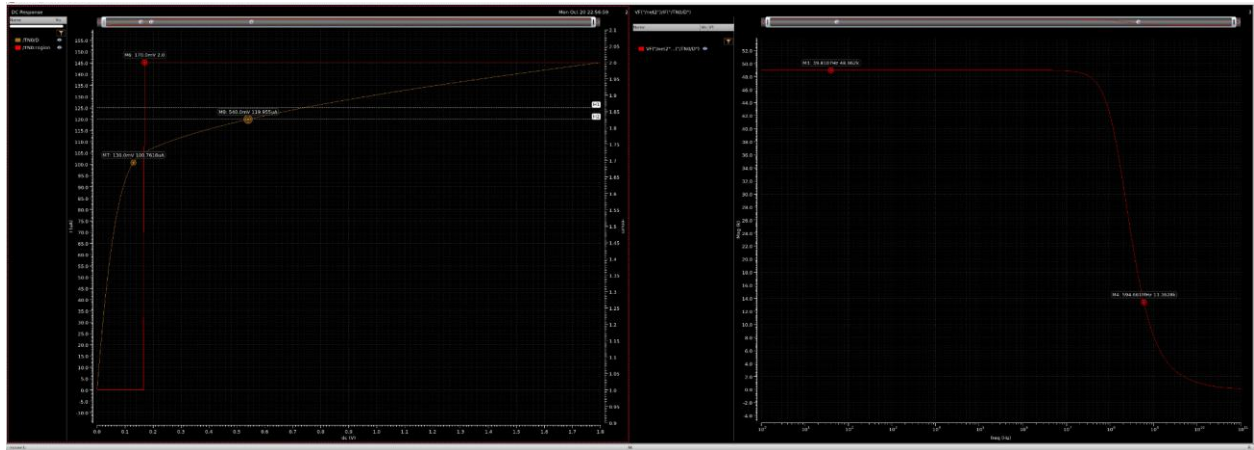
VI. SIMULATION RESULTS

Images of simulation results requested in the lab manual should be included here. Plots should be annotated to show relevant data points which will include either (1) data points that are explicitly requested in the lab manual or (2) data points from which other requested parameters are calculated. Data in the images should also be legible, so scale the images accordingly to make sure the data is easy to read. Post-layout simulations should be plotted in the same window with the pre-layout simulation results in order to make the differences between the two more apparent.

1.SIMPLE CURRENT MIRROR

Dc and ac analysis:





POST LAYOUT SIMULATION:



SIMULATION SETUP WINDOW:

ADE L (7) - Lab5 Current_mirrors_1_tb schematic@n01-zeus.olympus.ece.tamu.edu

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

27

Design Variables

Name	Value
p5vonly	0

Analyses

Type	Enable	Arguments
1 dc	☒	t 0 1.8 5m Linear Step Size Start-Stop /V0
2 ac	☒	1 100G 20 Logarithmic Points Per Decade Start-Stop

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 TN0/D		☐	☐	no
2 I3/I_OUT		☒	☒	yes
3 (VF("net1") / IF("I3/I_OUT"))	wave	☒	☐	

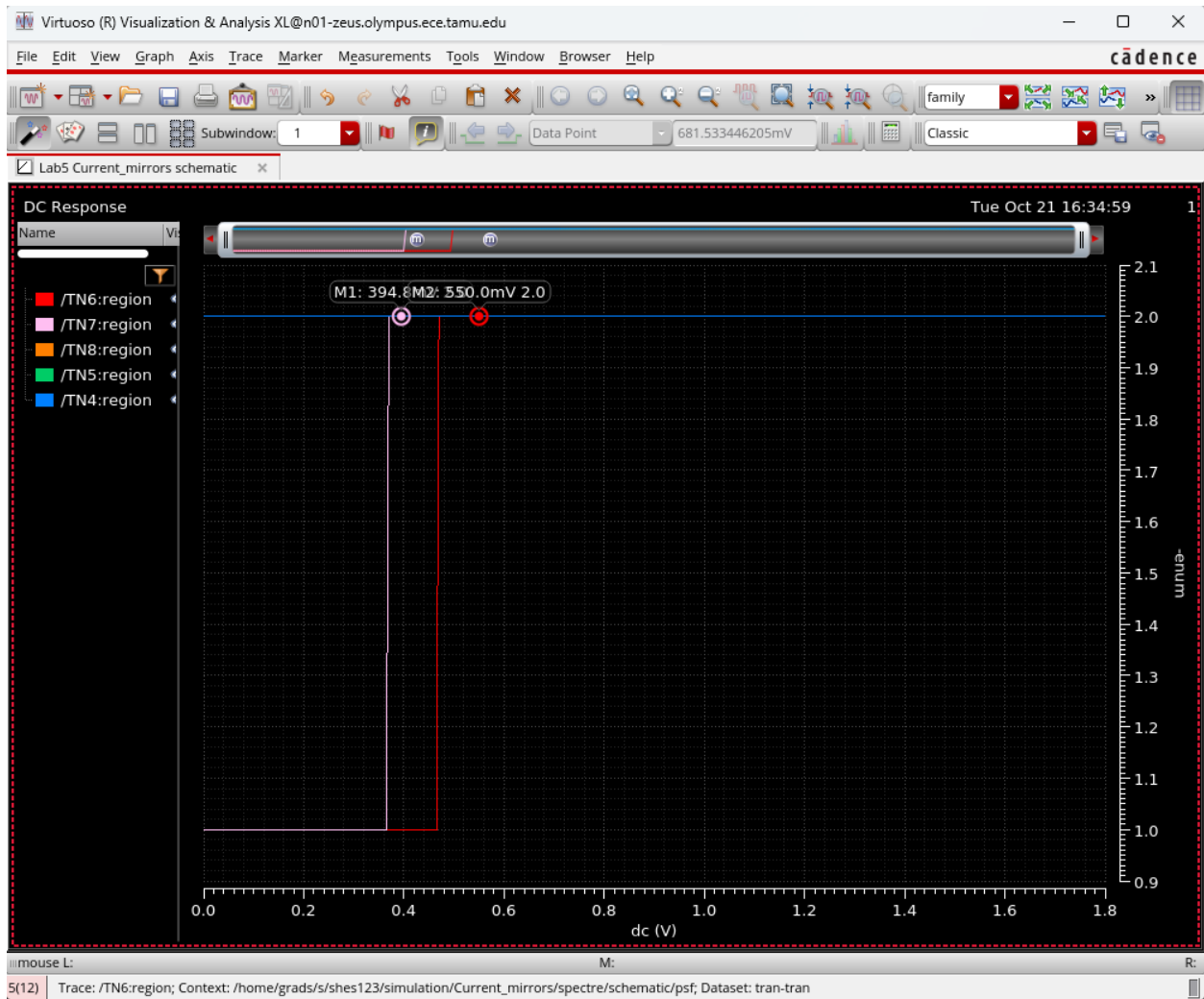
Plot after simulation: Auto Plotting mode: Append

> Results in ...shes123/simulation/Current_mirr

39(72) | Netlist and Run | Status: Ready | T=27 C | Simulator: spectre aps | State: spectre_state1

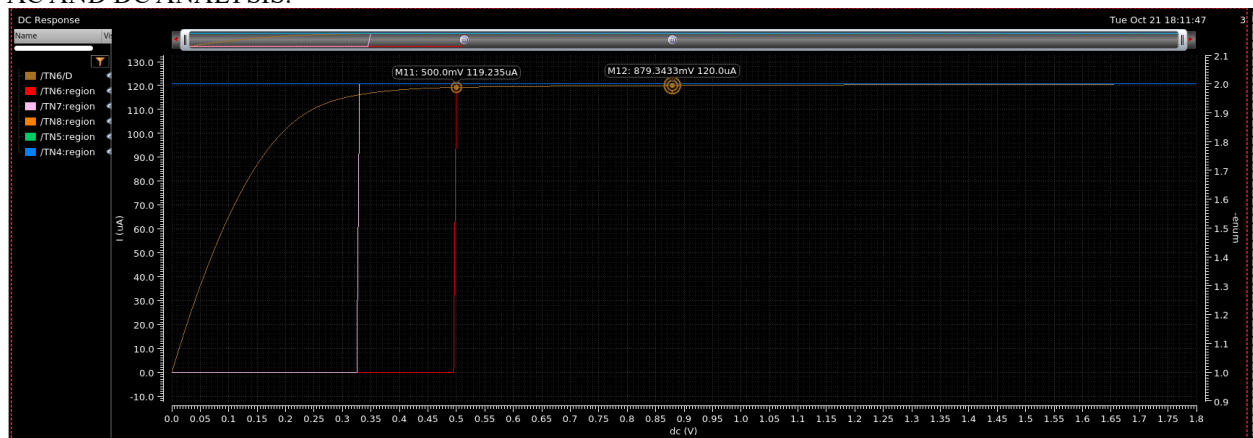
OPERATING REGIONS:

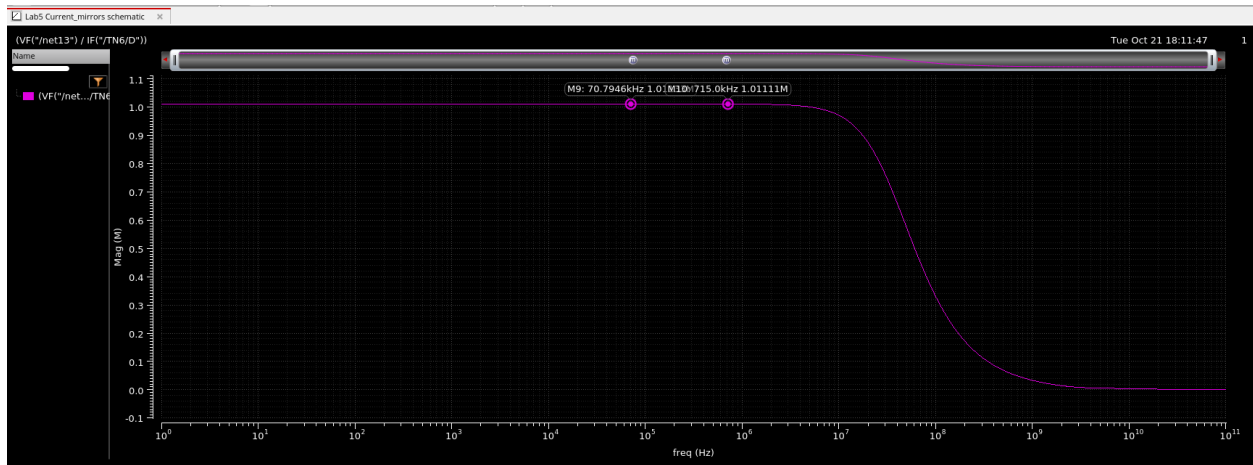




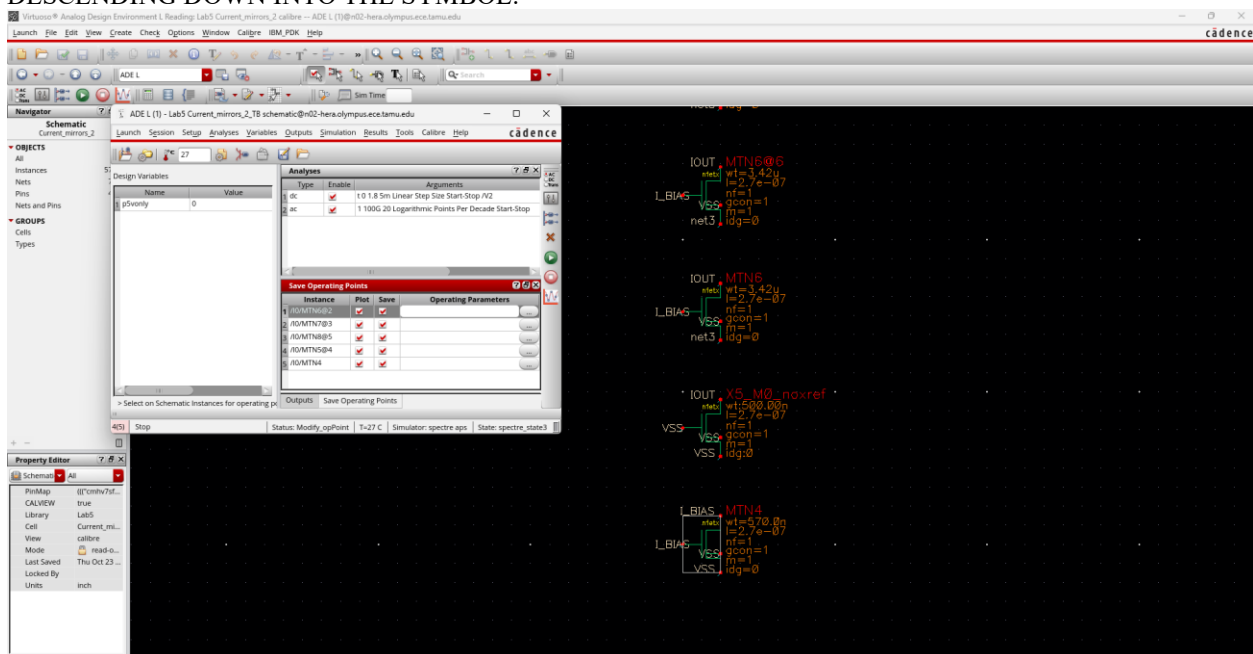
2.CASCODE CURRENT MIRROR

AC AND DC ANALYSIS:





DESCENDING DOWN INTO THE SYMBOL:



COMPARISON BETWEEN PRE-LAYOUT AND POST LAYOUT SIMULATION:

